Risk Assessment & Mitigation Strategy

Smart Emergency Response & Healthcare Coordination Platform

1. RISK ASSESSMENT MATRIX

ID	CATEGORY	RISK EVENT	PROB.	IMPACT	SEVERITY	PREVENTIVE ACTION	OWNER
R01	Technical	SignalR real-time connection fails, so live tracking and status updates are delayed	Medium	High	**HIGH**	Use SignalR automatic reconnect; build and test tracking early on its own; store every status change in the database so state can be reloaded.	Member 3
R02	Technical	Google Maps, GPS location, or Google Login fails, is denied by the user, or goes over quota	Medium	Medium	**MEDIUM**	Add third-party services one at a time behind simple interfaces; cache the last known routes; monitor API quota; let the patient type an address when GPS is unavailable or denied.	Member 6 / Member 2
R03	Technical	AI triage suggests a wrong priority or department (under- or over-triage)	Medium	High	**HIGH**	AI only suggests, with a confidence level; the operator confirms every case; no medical diagnosis; test against sample emergencies.	Team Lead
R04	Technical	Smart matching recommends an unsuitable ambulance or hospital because availability, capacity, or location data is missing or outdated	Medium	High	**HIGH**	Test the matching rules on sample cases before joining the main flow; keep availability, capacity, and location fields required and easy to update; the responsible staff member confirms every recommendation and can override it.	Team Lead / Member 4
R05	Security	Patient or medical data is exposed to the wrong role or emergency contact, or appears in logs and notifications	Medium	High	**HIGH**	Enforce role-based authorization with JWT and validate permissions on the backend APIs, not only in the frontend; use HTTPS; share only what each role needs; keep sensitive data out of logs and notifications; test permissions per role.	Member 2
R06	Schedule	Deadlines slip because of scope creep or work piling up near the end	High	High	**CRITICAL**	Stick to the MVP and the four milestones; work in short sprints; keep the basic versions of AI triage, matching, and tracking in scope and move advanced improvements to future work.	Team Lead
R07	Team	Unplanned absence, late delivery, or uneven workload between members	Medium	Medium	**MEDIUM**	Give every task one owner and a deadline; document the code; require peer review before each PR merge.	Team Lead
R08	Integration	Parts built by different members do not work together (API or data mismatch)	Medium	Medium	**MEDIUM**	Agree API contracts in Swagger early; merge small PRs into develop often; run integration tests before each milestone.	Team Lead

Severity is calculated from probability and impact (Low = 1, Medium = 2, High = 3). Score 9 = CRITICAL, 6 = HIGH, 3-4 = MEDIUM, 1-2 = LOW.

2. MITIGATION & CONTINGENCY PLANS (PLAN B)

ID	RISK DESCRIPTION	MITIGATION STRATEGY (IMMEDIATE ACTION)	CONTINGENCY PLAN (PLAN B)
R01	SignalR failure	SignalR falls back to long polling automatically; the app reloads the latest status from the API.	Operator and driver update statuses from their dashboards; the screens refresh from the API every few seconds.
R02	Maps, GPS, or Google Login issue	Show the last cached route; let the patient type an address if GPS is unavailable or permission is denied.	Operator confirms the location with the patient and enters it manually; email and password login stays available if Google Login fails.
R03	Wrong AI suggestion	Show every suggestion with its confidence level; the operator reviews each case before confirming.	Turn off AI suggestions and let the operator choose emergency type and priority from fixed lists.
R04	Wrong matching	The responsible staff member chooses another ambulance or hospital from the full list.	Manual dispatch from a list filtered by availability, distance, and department.
R05	Data exposure	Check endpoint permissions; remove sensitive data from logs and notifications; limit what emergency contact alerts contain.	Limit medical profile access to hospital staff and doctors until the issue is fixed; send emergency contacts status-only alerts; rotate JWT keys.
R06	Schedule delay	Move advanced improvements (more sophisticated AI, optimized matching, extensive analytics, advanced tracking) to future work.	Keep the basic AI triage, matching, and tracking in the core delivery; reassign members to MVP blockers and finish the core emergency workflow first.
R07	Team absence	Reassign open tasks and pull requests during the next team sync.	A back-up member takes over each area, based on the written documentation.
R08	Integration mismatch	Fix the API contract in Swagger and update both sides in the same sprint.	Team Lead runs a dedicated integration session and resolves breaking changes before merge.

3. GOVERNANCE & RISK MONITORING LIFECYCLE

Risk tracking is part of the team's normal GitHub workflow. To keep the risks under control until the final delivery, the team follows these rules:

- **Bi-Weekly Risk Review:** the Team Lead reviews the status of all HIGH and CRITICAL risks every two weeks.
- **Issue Reporting:** any new technical, security, or team risk is logged as a GitHub Issue with the label `kind/risk`.
- **PR Impact Check:** pull requests that change critical paths (emergency requests, ambulance and hospital matching, authentication, and notifications) are checked against this matrix before merge.

Approval & Sign-Off
This document was prepared according to the project submission requirements and is pending final review and merge into the `develop` branch by the Team Lead.